

Supplementary Information: Exact architecture exploration for heterogeneous edge–cloud cyber-physical systems

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Abstract

Supplementary Notes 1–2 and Supplementary Tables S1–S13 accompanying the main manuscript. All tabulated values are generated from the same result files as the main-text summaries.

1 Supplementary Note 1: correctness of the pruning rules

1.1 Setting

The design space is the Cartesian product $\mathcal{X} = D_1 \times \dots \times D_k$ traversed under a fixed variable order $v_1, \dots, v_k$. A *partial design* of depth d is a prefix of assignments $p = (a_1, \dots, a_d)$, $a_i \in D_{v_i}$, and its completions are $\mathcal{C}(p) = \{x \in \mathcal{X} : x_{v_i} = a_i, i \leq d\}$.

Let $\sigma : \mathcal{X} \rightarrow \{0, 1\}$ be the well-formedness predicate, $\mathcal{X}_\sigma$ the well-formed space, $\mathcal{F}$ the feasible set and $\mathcal{F}_\sigma = \mathcal{F} \cap \mathcal{X}_\sigma$. Objectives $F(x) = (f_1(x), \dots, f_n(x))$ are minimised, $u \preceq v$ denotes $u_i \leq v_i$ for all i with at least one strict inequality, and

$$\mathcal{P}^* = \{F(x) : x \in \mathcal{F}_\sigma, \nexists y \in \mathcal{F}_\sigma : F(y) \preceq F(x)\}. \quad (\text{S1})$$

A bound LB_i is *admissible* for f_i if $LB_i(p) \leq f_i(x)$ for every partial design p and every $x \in \mathcal{C}(p)$.

1.2 Construction of the bounds

Supplementary Theorem 1 (Term-wise relaxation) *Let $f_i(x) = \sum_{t \in T_i} \varphi_t(x)$ with $\varphi_t \geq 0$ and $\underline{\varphi}_t(p) = \min_{x \in \mathcal{C}(p)} \varphi_t(x)$. Then $LB_i(p) = \sum_{t \in T_i} \underline{\varphi}_t(p)$ is admissible.*

Proof For every $x \in \mathcal{C}(p)$ and every $t \in T_i$ we have $\underline{\varphi}_t(p) \leq \varphi_t(x)$, because the minimum is taken over a set that contains x . Summing over $t \in T_i$ gives $LB_i(p) \leq \sum_{t \in T_i} \varphi_t(x) = f_i(x)$. $\square$

The minima are taken independently per term, so the combination of values realising them need not correspond to any single completion; the bound is optimistic but valid. Every term of the model is monotone in each of its free variables, so each minimum is attained at a domain endpoint and is read from a table of precomputed extrema in constant time.

Supplementary Theorem 2 (Exactness on determined terms) *Let $\text{supp}(\varphi_t)$ be the set of variables φ_t depends on. If $\text{supp}(\varphi_t) \subseteq \{v_1, \dots, v_d\}$, then φ_t is constant on $\mathcal{C}(p)$, and substituting its exact value for $\underline{\varphi}_t(p)$ preserves admissibility while tightening the bound.*

Proof Constancy follows from the definition of $\mathcal{C}(p)$, and a constant equals its own minimum over $\mathcal{C}(p)$, so Theorem 1 applies with the substituted value. $\square$

Remark 1 (Role of the variable order) The order places the determinants of tier utilisation — node counts, node classes, replication factor and placement policy — before the remaining variables. Once they are assigned, the waiting term $\frac{w_i}{C_t} \cdot \frac{\rho_t}{2(1-\rho_t)}$ is constant on $\mathcal{C}(p)$ and enters the bound exactly by Theorem 2 instead of being relaxed to its trivial bound 0. This is what makes a contention-dependent objective usable for safe pruning without assignment monotonicity. If the determinants are fixed and $\rho_t \geq 1$, no completion has finite latency and $LB_L(p) = \infty$, which is admissible.

1.3 The three pruning rules

Supplementary Proposition 3 (Safe feasibility pruning) *If LB_{g_j} is admissible for the constrained quantity g_j and $LB_{g_j}(p) > g_j^{\max}$, then $\mathcal{C}(p) \cap \mathcal{F} = \emptyset$.*

Proof For any $x \in \mathcal{C}(p)$, admissibility gives $g_j(x) \geq LB_{g_j}(p) > g_j^{\max}$, so x violates the constraint and $x \notin \mathcal{F}$. $\square$

Supplementary Proposition 4 (Safe dominance pruning) *Let $y \in \mathcal{F}_\sigma$ be a fully evaluated feasible architecture with $F(y) \preceq LB(p)$. Then $F(x) \notin \mathcal{P}^*$ for every $x \in \mathcal{C}(p)$.*

Proof Let $x \in \mathcal{C}(p)$. Admissibility gives $f_i(x) \geq LB_i(p) \geq F_i(y)$ for all i , and by assumption there is a j with $F_j(y) < LB_j(p) \leq f_j(x)$. Hence y strictly dominates x , so $F(x)$ is not on the Pareto front. $\square$

Supplementary Proposition 5 (Closure of the well-formedness predicate) *Each invariant of σ depends only on assigned variables and, under the chosen order, all variables of an invariant are assigned before it is first checked. Since assignments are never revised, $\neg\sigma(p)$ implies $\sigma(x) = 0$ for all $x \in \mathcal{C}(p)$, that is $\mathcal{C}(p) \cap \mathcal{X}_\sigma = \emptyset$.*

Remark 2 The structural invariants define what counts as an architecture rather than following from the constraints of the problem: the evaluator can formally compute objectives for, say, a replicated deployment under a policy that places the ingest stage off the edge tier, although such a configuration is meaningless. Structural pruning is therefore part of the definition of the search space, and the exhaustive baseline applies the same predicate.

1.4 Front preservation

Supplementary Theorem 6 (Pareto-front preservation) *If all bounds are admissible, pruning is performed only by Propositions 3–5, and the archive contains only feasible fully evaluated architectures, then the set of non-dominated vectors returned by the exploration equals $\mathcal{P}^*$.*

Proof ($\subseteq$) Every returned vector is $F(x)$ for an evaluated $x \in \mathcal{F}_\sigma$ that survives non-dominated filtering. A vector dominating it would belong to an architecture that was either evaluated, in which case filtering would have removed the dominated one, or pruned; pruning by Proposition 3 or 5 would contradict its feasibility and well-formedness, and pruning by Proposition 4 implies a further feasible evaluated architecture dominating it, so the argument repeats along a strictly decreasing chain in a finite set.

($\supseteq$) Let $u \in \mathcal{P}^*$ and $x \in \mathcal{F}_\sigma$ with $F(x) = u$. The traversal is a prefix tree, so x is reached through the unique chain $p_0 \subset p_1 \subset \dots \subset p_k = x$. If some p_d were pruned, then by Proposition 5 $\sigma(x) = 0$ and by Proposition 3 $x \notin \mathcal{F}$, both contradicting $x \in \mathcal{F}_\sigma$; by Proposition 4 there would exist a feasible y strictly dominating x , contradicting $u \in \mathcal{P}^*$. Hence no p_d is pruned, x is evaluated, and u survives the final filtering. $\square$

Remark 3 (Duplicates) The theorem is stated in objective space. If two distinct architectures share an objective vector, dominance pruning may retain only one of them; the front is unaffected. All experiments therefore compare sets of objective vectors.

Remark 4 (Order independence) Proposition 4 holds for any archive of evaluated feasible architectures, so the traversal order affects pruning efficiency but never correctness. This is confirmed empirically in Supplementary Table S7, where all order and toggle combinations return the same front.

1.5 Complexity and break-even

With C_E the cost of one full evaluation, C_P the cost of computing the bounds at one node, N the number of expanded partial designs, N_E the number of full evaluations and $|A|$ the archive size,

$$T_{\text{exh}} = O(|\mathcal{X}| C_E), \quad T_{\text{HCA}} = O(N(C_P + n|A|) + N_E C_E), \quad (\text{S2})$$

so pruning is profitable when $C_E > N(C_P + n|A|)/(|\mathcal{X}| - N_E)$. The right-hand side is measured rather than assumed in the main text.

2 Supplementary Note 2: full evaluation model

With $\mathcal{H}(x)$ the hop set implied by the placement policy and the replication factor, Λ the system arrival rate, T the accounting horizon and ρ_t the tier utilisation defined in Methods:

$$E(x) = \sum_{t \in \{e, g\}} N_t \left(P_t^{\text{idle}} + \rho_t (P_t^{\text{peak}} - P_t^{\text{idle}}) \right) T + \rho_c P_c^{\text{peak}} T + \sum_{(s, \ell) \in \mathcal{H}(x)} s \Lambda e_\ell T, \quad (\text{S3})$$

$$C(x) = n_e (c_{\kappa_e} + c_\ell) + n_g (c_{\kappa_g} + c_\ell) + c_{\kappa_c}, \quad (\text{S4})$$

$$D(x) = \Lambda \sum_{(s, \ell) \in \mathcal{H}(x)} s. \quad (\text{S5})$$

An architecture is feasible when $\rho_t \leq 0.85$ on every used tier, the resident memory of the stages mapped to a tier fits the memory of its node class, the aggregate access-link traffic does not exceed the total bandwidth of the attached nodes, and the latency, cost and energy budgets hold. Energy is reported in kWh over $T = 1$ h. Cost coefficients are synthetic normalised values encoding relative price tiers; no monetary interpretation is claimed, and neither the energy nor the cost model is validated against measurements.

3 Supplementary Tables

The main text compares analytical predictions with the mean latency measured over each deployment run. Supplementary Table S4 is the complementary robustness analysis, in which the same predictions are compared with the measured median (p_{50}) of the same runs.

Table S1 Full budget-matched comparison on S2. Medians over 20 seeds.

scale	workload	method	evals	recall	HV ratio	IGD ⁺	time [s]
S2	telemetry	exhaustive	3 888	1.000	1.000	0.0000	0.044
		random ($B=100$)	100	0.037	0.642	0.1455	0.001
		random ($B=250$)	250	0.074	0.795	0.1033	0.003
		random ($B=500$)	500	0.148	0.882	0.0602	0.006
		random ($B=1000$)	1000	0.259	0.927	0.0388	0.014
		NSGA-II ($B=100$)	100	0.018	0.580	0.1773	0.010
		NSGA-II ($B=250$)	250	0.241	0.883	0.0467	0.028
		NSGA-II ($B=500$)	500	0.685	0.981	0.0081	0.064
		NSGA-II ($B=1000$)	1000	0.926	0.999	0.0002	0.147
		HCA-DSE	44	1.000	1.000	0.0000	0.036
S2	control	exhaustive	3 888	1.000	1.000	0.0000	0.042
		random ($B=100$)	100	0.040	0.652	0.1598	0.001
		random ($B=250$)	250	0.080	0.791	0.1056	0.003
		random ($B=500$)	500	0.120	0.875	0.0556	0.006
		random ($B=1000$)	1000	0.240	0.922	0.0377	0.014
		NSGA-II ($B=100$)	100	0.020	0.594	0.1519	0.011
		NSGA-II ($B=250$)	250	0.320	0.902	0.0373	0.030
		NSGA-II ($B=500$)	500	0.720	0.993	0.0065	0.070
		NSGA-II ($B=1000$)	1000	0.800	0.999	0.0030	0.160
		HCA-DSE	37	1.000	1.000	0.0000	0.035
S2	sensing	exhaustive	3 888	1.000	1.000	0.0000	0.029
		random ($B=100$)	100	0.031	0.359	0.2624	0.001
		random ($B=250$)	250	0.062	0.750	0.1344	0.002
		random ($B=500$)	500	0.156	0.849	0.0859	0.005
		random ($B=1000$)	1000	0.250	0.897	0.0626	0.010
		NSGA-II ($B=100$)	100	0.000	0.271	0.3183	0.010
		NSGA-II ($B=250$)	250	0.250	0.725	0.0983	0.030
		NSGA-II ($B=500$)	500	0.625	0.941	0.0172	0.073
		NSGA-II ($B=1000$)	1000	0.875	1.000	0.0017	0.182
		HCA-DSE	26	1.000	1.000	0.0000	0.016

Table S2 Full budget-matched comparison on S3. Medians over 20 seeds.

scale	workload	method	evals	recall	HV ratio	IGD ⁺	time [s]
S3	telemetry	exhaustive	34 560	1.000	1.000	0.0000	0.308
		random ($B=100$)	100	0.000	0.372	0.3783	0.001
		random ($B=250$)	250	0.000	0.546	0.2558	0.002
		random ($B=500$)	500	0.000	0.613	0.1827	0.005
		random ($B=1000$)	1000	0.032	0.732	0.1147	0.011
		NSGA-II ($B=100$)	100	0.000	0.072	0.4807	0.010
		NSGA-II ($B=250$)	250	0.032	0.324	0.2723	0.026
		NSGA-II ($B=500$)	500	0.258	0.763	0.1132	0.061
		NSGA-II ($B=1000$)	1000	0.500	0.946	0.0454	0.145
		HCA-DSE	50	1.000	1.000	0.0000	0.138
S3	control	exhaustive	34 560	1.000	1.000	0.0000	0.265
		random ($B=100$)	100	0.000	0.365	0.4042	0.001
		random ($B=250$)	250	0.000	0.525	0.2933	0.002
		random ($B=500$)	500	0.000	0.609	0.1771	0.005
		random ($B=1000$)	1000	0.000	0.725	0.1189	0.010
		NSGA-II ($B=100$)	100	0.000	0.114	0.4329	0.010
		NSGA-II ($B=250$)	250	0.037	0.575	0.2181	0.027
		NSGA-II ($B=500$)	500	0.352	0.888	0.0427	0.064
		NSGA-II ($B=1000$)	1000	0.648	0.984	0.0236	0.147
		HCA-DSE	41	1.000	1.000	0.0000	0.125
S3	sensing	exhaustive	34 560	1.000	1.000	0.0000	0.266
		random ($B=100$)	100	0.000	0.045	0.5753	0.001
		random ($B=250$)	250	0.000	0.311	0.3251	0.002
		random ($B=500$)	500	0.000	0.446	0.2306	0.005
		random ($B=1000$)	1000	0.000	0.658	0.1569	0.010
		NSGA-II ($B=100$)	100	0.000	0.030	0.6768	0.010
		NSGA-II ($B=250$)	250	0.018	0.126	0.3325	0.027
		NSGA-II ($B=500$)	500	0.204	0.713	0.2267	0.064
		NSGA-II ($B=1000$)	1000	0.741	0.951	0.1677	0.175
		HCA-DSE	43	1.000	1.000	0.0000	0.085

Table S3 Full S3 statistical comparison: one-sided Mann–Whitney test for larger NSGA-II hypervolume and Cliff’s δ . Unadjusted p -values, exploratory.

workload	budget	NSGA-II HV med.	random HV med.	p	Cliff’s δ	NSGA-II recall	HCA-DSE evals
telemetry	100	0.0721	0.3723	0.999	-0.565	0.0	50
telemetry	250	0.3235	0.546	0.938	-0.282	0.032	50
telemetry	500	0.7628	0.6125	0.338	0.08	0.258	50
telemetry	1000	0.9455	0.7322	0.299	0.1	0.5	50
control	100	0.1144	0.3648	0.975	-0.36	0.0	41
control	250	0.5746	0.5252	0.430	0.035	0.037	41
control	500	0.8884	0.6087	0.090	0.25	0.352	41
control	1000	0.9835	0.7246	0.057	0.295	0.648	41
sensing	100	0.0299	0.0449	0.955	-0.31	0.0	43
sensing	250	0.1262	0.3109	0.780	-0.14	0.018	43
sensing	500	0.7127	0.4465	0.308	0.095	0.204	43
sensing	1000	0.9506	0.6581	0.142	0.2	0.741	43

Table S4 Testbed validation against measured p_{50} , including utilisation and reproducibility subsets. Pair agreement is a fraction; undefined values (no eligible pairs) are shown as –.

subset	n	MAPE [%]	med. AE [ms]	Spearman ρ	Kendall τ	pair > 10%	pair > 25%	repeat IQR [ms]
telemetry (all)	12	19.9	7.43	0.811	0.697	0.815	0.895	0.23
telemetry ($\rho \leq 0.75$)	6	15.4	2.85	0.829	0.733	0.818	1.0	0.2
control (all)	12	14.4	2.49	0.865	0.748	0.896	0.909	0.11
control ($\rho \leq 0.75$)	10	9.4	2.32	0.766	0.629	0.815	0.833	0.11
sensing (all)	12	51.1	19.15	0.935	0.84	0.935	0.913	1.0
sensing ($\rho \leq 0.75$)	10	51.0	18.82	0.985	0.944	1.0	1.0	0.75
sensing (reproducible)	8	39.3	18.56	0.97	0.909	1.0	–	0.67
all workloads (all)	36	28.4	7.43	0.968	0.884	0.961	0.98	0.27
all workloads ($\rho \leq 0.75$)	26	26.8	3.96	0.945	0.855	0.959	0.982	0.22
all workloads (reproducible)	32	22.7	4.5	0.957	0.869	0.956	0.979	0.23

Table S5 Exact finite-table SMT baseline. Solver time includes encoding; total time also includes full table evaluation. “Calls” counts full evaluator invocations.

scale	workload	calls	feasible	table [s]	solver [s]	total [s]	front
S1	telemetry	1080	494	0.006	0.332	0.338	17
S1	control	1080	520	0.007	0.424	0.431	18
S1	sensing	1080	228	0.006	0.160	0.166	11
S2	telemetry	3888	1743	0.023	3.297	3.319	27
S2	control	3888	1753	0.022	2.209	2.231	25
S2	sensing	3888	910	0.023	0.929	0.951	16

Table S6 Ablation on S3: expanded partial designs, full evaluations and runtime for each workload.

variant	telemetry			control			sensing		
	states	evals	[ms]	states	evals	[ms]	states	evals	[ms]
structural only	56 797	34 560	322	56 797	34 560	294	56 797	34 560	282
structural + feasibility bounds	22 205	12 700	490	16 110	7 025	256	21 084	10 956	280
structural + dominance bounds	15 552	322	610	15 648	407	354	7 961	415	160
HCA-DSE (all layers)	5 828	50	134	5 826	41	127	4 393	43	99

Table S7 Full order and queue-bound ablation on S1–S3. All configurations preserve the unrounded exhaustive front.

scale	workload	order	queue	states	checks	evals	active	pruned	ms
S1	telemetry	O1	off	695	1299	136	0	0	9.7
S1	telemetry	O1	on	589	1299	30	0	0	10.6
S1	telemetry	O2	off	447	746	136	0	0	5.8
S1	telemetry	O2	on	249	516	32	306	46	4.8
S1	control	O1	off	679	1317	112	0	0	9.5
S1	control	O1	on	593	1317	26	0	0	10.5
S1	control	O2	off	391	669	113	0	0	5.3
S1	control	O2	on	242	512	28	312	35	4.7
S1	sensing	O1	off	497	819	148	0	0	5.8
S1	sensing	O1	on	365	819	16	0	0	5.7
S1	sensing	O2	off	481	810	147	0	0	5.9
S1	sensing	O2	on	226	503	16	309	72	3.7
S2	telemetry	O1	off	1816	3940	208	0	0	30.6
S2	telemetry	O1	on	1652	3940	44	0	0	33.4
S2	telemetry	O2	off	939	1815	208	0	0	15.1
S2	telemetry	O2	on	604	1383	46	839	99	13.0
S2	control	O1	off	1780	3960	164	0	0	29.9
S2	control	O1	on	1653	3960	37	0	0	34.2
S2	control	O2	off	835	1674	165	0	0	13.1
S2	control	O2	on	587	1364	39	842	80	12.0
S2	sensing	O1	off	1268	2421	290	0	0	17.5
S2	sensing	O1	on	1004	2421	26	0	0	15.8
S2	sensing	O2	off	1161	2184	289	0	0	15.1
S2	sensing	O2	on	600	1427	26	879	177	9.9
S3	telemetry	O1	off	6029	14794	251	0	0	113.0
S3	telemetry	O1	on	5828	14794	50	0	0	132.5
S3	telemetry	O2	off	2871	6806	251	0	0	51.2
S3	telemetry	O2	on	2397	6100	52	3777	188	53.4
S3	control	O1	off	5990	14812	205	0	0	108.6
S3	control	O1	on	5826	14812	41	0	0	129.0
S3	control	O2	off	2755	6636	206	0	0	48.2
S3	control	O2	on	2375	6074	43	3777	163	50.9
S3	sensing	O1	off	4888	11367	538	0	0	83.9
S3	sensing	O1	on	4393	11367	43	0	0	83.2
S3	sensing	O2	off	4351	9905	537	0	0	72.5
S3	sensing	O2	on	3221	8281	43	5139	419	60.8

Table S8 Design variables and domain sizes of the four design-space scales.

variable	meaning	S1	S2	S3	S4
n_e	number of edge nodes	3	6	24	80
$edge_cls$	edge node class	3	3	3	3
n_g	number of gateways	2	3	5	8
gw_cls	gateway class	2	2	2	2
$link_cls$	access link class	3	3	3	3
$cloud_cls$	cloud instance class	2	2	2	2
rep	replication factor of stage 1	2	3	4	5
$policy$	placement policy	3	3	3	3
design-space size $ \mathcal{X} $		1 296	5 832	51 840	345 600

Table S9 Workload classes. Stage values are given as $\tau_1/\tau_2/\tau_3$.

workload	n_d	λ	Λ	w_i [MI]	s_i [kB]	D [ms]	$C^{\max}$
telemetry	200	2	400	8/4/20	20/10/10	120	60
control	60	10	600	5/3/9	5/4/3	30	60
sensing	40	1.5	60	120/30/260	1200/350/300	900	90

Table S10 Resource classes used in all experiments.

class	C / B	M / τ	P^{idle} / e	P^{peak}	cost
<i>compute node classes</i>					
edge-low	2000	2048	4	12	1
edge-med	6000	8192	8	30	2.4
edge-high	15000	16384	15	65	5.2
gw-low	4000	4096	6	20	1.8
gw-high	12000	16384	12	48	4
cloud-med	40000	131072	0	180	6
cloud-high	100000	262144	0	380	12
<i>link classes (name, B [Mb/s], τ [ms], e [J/MB], cost)</i>					
constrained	10	4	0.3	0.2	–
standard	100	2	0.12	0.6	–
fast	1000	1	0.05	1.4	–

Table S11 Exploration funnel across all scales and workloads.

scale	workload	$ \mathcal{X} $	structural pruned	feasibility pruned	dominance pruned	full evals	feasible	$ \mathcal{P} $	full-eval reduction [%]	exhaustive evals	recall
S1	telemetry	1 296	138	233	895	30	23	17	97.69	1 080	1.0
S1	control	1 296	138	269	863	26	23	18	97.99	1 080	1.0
S1	sensing	1 296	77	613	590	16	11	11	98.77	1 080	1.0
S2	telemetry	5 832	1 422	1 127	3 239	44	36	27	99.25	3 888	1.0
S2	control	5 832	1 425	1 236	3 134	37	34	25	99.37	3 888	1.0
S2	sensing	5 832	1 075	2 323	2 408	26	19	16	99.55	3 888	1.0
S3	telemetry	51 840	8 374	27 573	15 843	50	42	31	99.90	34 560	1.0
S3	control	51 840	8 385	28 512	14 902	41	38	27	99.92	34 560	1.0
S3	sensing	51 840	7 176	27 328	17 293	43	36	27	99.92	34 560	1.0
S4	telemetry	345 600	24 331	292 656	28 558	55	47	32	99.98	–	–
S4	control	345 600	24 351	294 344	26 854	51	48	29	99.99	–	–
S4	sensing	345 600	23 497	286 394	35 657	52	45	28	99.98	–	–

Table S12 Scalability sweep: expanded partial designs, full evaluations and runtime.

$ \mathcal{X} $	telemetry			control			sensing		
	states	evals	[s]	states	evals	[s]	states	evals	[s]
1.3×10^3	656	40	0.12	659	32	0.11	391	18	0.05
5.8×10^3	1 989	52	0.41	1 997	52	0.38	1 142	30	0.18
5.2×10^4	7 286	55	1.61	7 293	55	1.49	5 183	53	0.94
3.5×10^5	10 341	56	2.31	10 348	56	2.12	8 643	59	1.64
2.8×10^6	14 076	56	3.14	14 083	56	2.97	12 056	61	2.41
1.6×10^7	15 613	56	3.44	15 620	56	3.55	13 216	61	2.71
8.3×10^7	16 773	56	3.98	16 780	56	3.66	13 888	61	3.11

Table S13 Analytical mean latency versus the discrete-event evaluator: 40 designs per workload, five seeds per design.

workload	designs	MAPE [%]	Spearman ρ
telemetry	40	39.3	0.921
control	40	44.3	0.919
sensing	36	14.2	0.963